

Part 2

Preparing the shared Blender model for TF3

What to add to, remove from and rename in the TSC-structured scene so the Model Editor imports it cleanly: axes, origins, node and material names, LOD files, PBR textures, locators. Nothing here breaks the TSC export.

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Import rules are from the Model Editor page (“FBX Model Import”, “Nodes / Meshes”, “Materials”, “Textures”, “Metadata”). The automatic mapping described is what `tools/blender/tf3_export_prep.py` performs.

1 · Three habits that make the model dual-target

- 1
- Every moving or physically meaningful part is its own object with its origin on its axis. TSC merges static parts anyway; TF3 needs axles as separate nodes (“the origin should be aligned in the center in terms of the x and z axis. Otherwise the axle will wobble ingame”).

manual

 Wheels modelled as one mesh per bogie look fine in TSC and are unusable in TF3.
- 2
- Keep the TSC naming as master, and never rely on Blender suffixes. The TF3 importer strips #... , and the add-on turns 1_0500_bo01wh02 into w2 ; a stray .001 would become part of a TSC node name and a TF3 mesh file name.
- 3
- Place a driver empty (tsc_driver) and keep the pantograph, doors and wipers as independent actions. TSC uses the empty for DriverPosition ; TF3 turns it into a crew seat locator. Independent actions become TSC .lia files and TF3 event strips without re-keying.

2 · Axes and the rotation at the branch

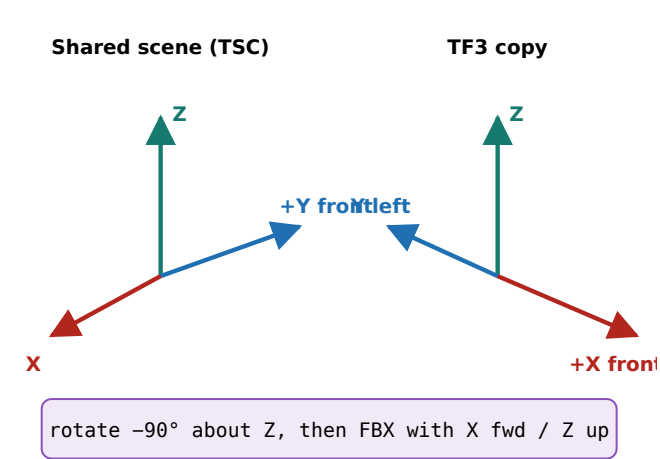


Figure 2.1 · The one geometric change at the branch. TF3 measures positions “along the lane” on x and places seats at x = ±8.5 on a loco

manual

, so the front must point along +X. The add-on applies the rotation to the temporary copy only.

- Export FBX with **Forward = X, Up = Z** and tick **Use Blender Coordinate System** in the Model Editor's Im-/Export options.

manual
- Scale: metres in both games. Apply scale before export (TSC rule; FBX exports with apply_scale_options = FBX_SCALE_ALL).
- Origin: the vehicle centre on the rail head in both games.

check

 that TF3's ground plane (“gray ground face at height 0” in the Model Editor) is the rail head for rail vehicles, not the sleeper top.
- Axle rotation axis: TF3 spins axles about their local y axis. After the -90° rotation, the TSC wheel objects (which spin about Blender X) rotate about TF3 y. The add-on rotates whole root objects, so children keep their relative orientation.

3 · Node names: the TSC → TF3 map

TSC object	TF3 node	Rule
1_0500_engine (main body)	body	Base names engine , wagon , coach , body , vehicle → body .
1_0500_bo01 / bo02	bogie_front / bogie_rear	Any node holding axle nodes is a bogie; the name is free. manual

TSC object	TF3 node	Rule
1_0500_bo01wh01 ... bo02wh03	w1 ... w6	Numbered front to back by position; listed in railVehicle.config.axles .
2_1200_engine (LOD twin)	body in ..._lod1.fbx	Same node name, next LOD file.
1_0128_panto01_part01..03	panto01_part01..03	Prefix stripped; chain kept; animation becomes an event strip (Part 3).
1_0064_wiper01_l	wiper01_l	Prefix stripped.
tsc_driver (empty)	seat_driver seat_crew_driving_upright	Imported as a crew seat. manual
emitter_smoke_dark (empty)	unchanged	“emitter” in the name → particle emitter; smoke_dark sets defaults; X scale = velocity. manual
Any object with tf3_name property	that value	Manual override.
1_0400_shadow_* , 1_1000_primarydigits_* , 1_1000_light_* , cab objects, tsc_coupling_*	(skipped)	No TF3 equivalent; see §9. Mark anything else with tf3_skip .

Why not name things the TF3 way from the start?

Because TSC's keyword names (bo01 , bo01wh01 , panto01 , wiper01) are what BRIAGE uses to keep parts un-merged and what the TSC bogie blueprint references. TF3 has no reserved names at all, so it is the flexible side. Keep the strict naming as master and generate the lax one.

4 • LODs: from name prefixes to files

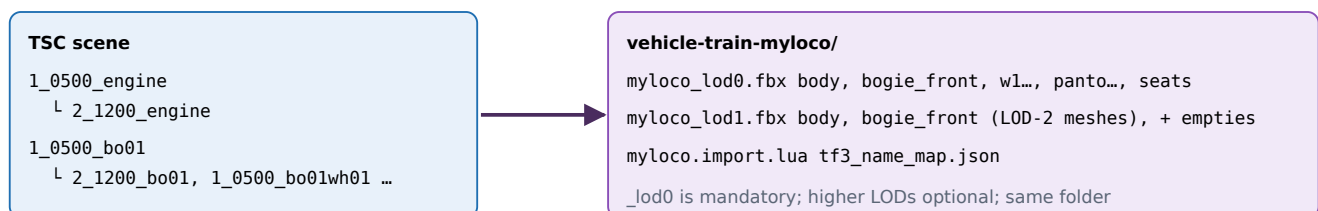


Figure 2.2 • LOD split. “Each .fbx file may contain the 3D model of one level of detail... _lodX.fbx ending, where X=0 is mandatory” **manual**. The add-on puts each TSC LOD level into its own file and carries non-mesh objects (seats, emitters) into every LOD so the hierarchy exists there too.

- Visible distances are set per LOD in the Model Editor (visibleFrom / visibleTo), not in names. **manual** Your TSC distances (500, 1200) are a good starting point.

- A node that exists in LOD 0 but not in LOD 1 simply disappears at distance. Animated parts (doors, pantograph) usually need at least a crude LOD 1 version; in TSC they often have none.
- Fake bogies must be listed once per LOD (`fakeBogies = { {...}, {...} }`) even if empty. [manual](#)

5 • Bogies, axles and the origin rules

Part	Origin	Parent	TSC equivalent
Axle (<code>w1</code> ...)	Axle centre; centred in x and z of the wheelset; spins about local y. manual	Its bogie	<code>bo01wh01</code> , radius typed into the bogie bp
Bogie (<code>bogie_front</code>)	Midway between its axles, at pivot height; the game pivots here. manual	body	<code>bo01</code> , pivot typed as <code>BogeyPivotX</code>
Body (<code>body</code>)	Vehicle centre on the rail.	RootNode	<code>engine</code>
Single-truck car (streetcar)	One bogie node under <code>body</code> ; add a fake bogie on <code>body</code> so there is a second alignment point.	-	One bogie at <code>BogeyPivotX = 0</code>
Steam coupled wheels	Each driving axle its own node under a frame node that is a child of <code>body</code> ; rods driven by the <code>drive / wheels</code> events (Part 3).	frame	TSC uses custom keywords to keep rods separate

The TSC toolkit's **Validate** already checks wheel bottoms on `Z = 0` and bogie origins; both checks are exactly what TF3 needs, so run it before a TF3 export too.

6 • Materials and the shortcode tag

In the FBX, a material name is `<materialname>[<shortcode>][#<comment>]` . The shortcode selects the TF3 material type; without it the importer predicts a type (or your `.import.lua` 's `predictMaterial` does). [manual](#) Names starting with `/` reference existing materials (e.g.

`::/vehicle/train/emissive/train_all_lights.mtl` for lights). [manual](#)

TSC material (shader)	TF3 material type (from manual names)	Note
Body, Underside — <code>TrainBasicObjectSpecular.fx</code>	physical with normal map + colour blend (the manual names <code>phys_transparent_nrml_map_cblend.mat</code> and <code>::physical_nrml_map_op_uv1_ao_op_mask.mat.lua</code>)	Shortcode strings are in the material-type table not supplied check . Set via material property <code>tf3_material</code> .
Transparency — <code>Lit3DirSingle</code> , alpha blend	PHYS_TRANSPARENT (placeholder/glass)	Window glass.
Lights — <code>UnlitSingle</code> , additive	reference <code>::/vehicle/train/emissive/train_all_lights.mtl</code>	Reuse the base-game emissive material instead of making one. manual
Number digits — <code>TrainDecal.fx</code>	(drop) → TF3 labels	See §8.

TSC material (shader)	TF3 material type (from manual names)	Note
<code>shadow_*</code> — <code>StencilShadow.fx</code>	(drop)	TF3 computes shadows.

Keep material *count* low here too: each group of a mesh with a different material becomes a sub-mesh, and “the number of submeshes needs to match the number of assigned materials” or the Model Editor patches and warns. [manual](#)

7 · Textures: diffuse/spec → PBR

The importer finds textures by the material name plus suffix, or by assignment in the FBX. [manual](#)

Map	Suffix	Content	From the TSC set
<code>map_albedo</code> / <code>_opacity</code>	<code>_alb</code> / <code>_albo</code>	Colour (+ alpha)	TSC diffuse, with baked shading removed
<code>map_normal</code>	<code>_nrm</code>	Normal map	TSC normal if the shader had one, else bake or flat
<code>map_metal_gloss_ao</code>	<code>_mga</code>	R metal, G gloss, B AO (TF2 convention check)	TSC specular → gloss channel; AO from a bake
<code>map_cblend_dirt_rust</code>	<code>_cdr</code>	Recolourable areas, dirt, rust masks	New: paint the livery mask
<code>map_lgt</code>	<code>_e</code>	Emissive	TSC night textures
<code>map_ao</code>	<code>_uv1_ao</code>	Lightmap AO on UV1	Optional

- Work in `.tga`; the Model Editor converts to `.dds` on save and the manual warns against basing work on lossy DDS. [manual](#)
- Power-of-two sizes, as in TSC. The TF3 budget pages were not supplied [check](#).
- Part 7 includes a small channel-packing script to build `_mga` from separate metal/gloss/AO greyscale images.

8 · Locators: seats, emitters, lights, bounding box, labels

Blender object	What the importer does manual
Empty <code>name seat_<anim></code> / <code> seat_crew_<anim></code>	Seat with position/rotation/scale; <code>anim</code> ∈ <code>idle</code> , <code>sitting</code> , <code>driving_upright</code> , <code>driving</code> . Meshes on the node are removed.
Empty with <code>emitter</code> in the name	Particle emitter; <code>particleId</code> = name before <code>#</code> ; velocity from X scale; <code>smoke_dark</code> / <code>smoke_bright</code> set defaults.
Blender light (Point, Spot, Sun)	Light source: colour, strength (Power × 100), angles; <code>childId</code> = parent node.
Mesh <code>name bounding_box</code>	Its bounds become the model's bounding box; mesh removed. Also <code> extent</code> (simulation extent) and <code> collider_list</code> (box collider).

Blender object	What the importer does manual
Two nested empties <code>... label</code>	A text label rectangle (parent = bottom-left, child = top-right; same rotation, z forward). Replaces TSC's number digits.

The add-on adds a `bbox|bounding_box` cube around the LOD-0 meshes automatically and converts `tsc_driver`; add seats for passengers and labels by hand (they have no TSC counterpart).

9 · What to leave out

- **Cab interior and pick meshes:** TF3 has no cab view. Keep them in a separate TSC scene or tag `tf3_skip`.
- **Stencil shadow meshes** and the **collision IGS hull:** TF3 computes shadows and uses `|collider_list` instead.
- **Number digit planes:** use labels.
- **Coupler empties:** TF3 has no couplers to author; vehicle spacing comes from the bounding box / extent check.
- **Light glow meshes** can stay if you want visible lamps, re-materialled to the base-game emissive material and switched by the `..._parts_on/off` events.

10 · Export checklist

✓	Check	Who catches it
☐	TSC toolkit Validate passes (scale, origins, wheel contact, names)	TSC toolkit
☐	Every axle is its own object under its bogie; bogie origin midway between axles	Validate + eye
☐	Driver empty <code>tsc_driver</code> present; passenger seat empties if a coach	Preview TF3 names
☐	Materials tagged <code>tf3_material</code> ; textures named <code><material>_alb/_nrm/_mga/_cdr</code> in the Textures folder	Model Editor validation
☐	Animated parts tagged <code>tf3_event</code> with a valid event name (Part 3)	Preview TF3 names
☐	TSC-only objects skipped (<code>tf3_skip</code> or default list)	Preview TF3 names
☐	Export root is inside the mod's staging folder or a scratch folder you drag onto the Model Editor	You